

# M15 Competency Practical

Retail Orders - integrated cloud engineering proof

**ASSESSMENT | CREATE YOUR OWN EVIDENCE**

**Static/simulator success is not a live Fabric pass.**

# Practical contract - Retail Orders

## Scenario

You own a small Fabric-style data flow for Retail Orders. Use the bundled normal fixture, but do not start from a solved architecture.

## Required design

Source -> Bronze/raw -> Silver validated Delta -> Gold/serving -> SQL consumer. Include least privilege, orchestration, monitoring, environment separation and rollback.

## Runtime truth

You may complete static/simulator parts offline. Any Fabric item/run/permission claim remains pending until actually observed in your tenant.

## Part A - data contract and quality

### Do

State one-row grain, required fields, accepted/quarantine rules and raw preservation.

### Control

Reconcile 7 raw -> 4 accepted + 3 quarantined; accepted amount = 3300.

### Evidence

Save rule table, rejected-row reasons and independent reconciliation.

## Part B - incremental state

### Do

Apply O2 update and O8 insert using stable order\_id. Define a watermark/checkpoint that moves only after durable publish.

### Control

Final state = 5 rows / 4250. Exact rerun is a no-op.

### Failure injection

Describe a publish failure and prove the previous successful checkpoint remains durable.

## Part C - platform/security/operations

### Design

Workspace/items, OneLake paths, notebook/pipeline boundaries, SQL serving choice, minimum permissions, connection/secret handling.

### Operate

Define run/freshness/count/amount monitoring plus one alert, recovery action and capacity/cost signal.

### Lifecycle

Dev/test/prod configuration separation, reviewed Git path, post-deploy validation and rollback.

## Part D - evidence and defense

### Evidence index

For each claim mark LIVE FABRIC, OFFLINE PLAN or SIMULATOR. Include object/path/run/permission/control evidence where genuinely observed.

### Teach back

Defend why your design is reproducible, least-privilege, rerunnable, observable and recoverable.

### Critical blockers

Fabric claim without execution; embedded secret; unresolved data reconciliation; unsafe checkpoint; publish-after-failed-validation.

# Pass and next route

## Pass condition

Every required competency is demonstrated with evidence and zero unresolved critical failures. No numeric average shortcut.

## If weak

Preserve attempt -> targeted repair -> changed micro-proof -> validation -> reassessment.

## Next

M15 PASS -> M16. G06 remains locked until M15 + M16 + M17 evidence is complete.